

Technical University of Denmark

Written examination: 22nd May 2024, 9 AM — 1 PM.

Course name: Introduction to Machine Learning and Data Mining.

Course number: 02450.

Aids allowed: All aids permitted (closed internet).

Exam duration: 4 hours.

Weighting: The individual questions are weighted equally.

The exam is multiple choice. All questions have four possible answers marked by the letters A, B, C, and D as well as the answer “Don’t know” marked by the letter E. Correct answer gives 3 points, wrong answer gives -1 point, and “Don’t know” (E) gives 0 points.

This exam only allows for electronic hand-in.

You hand in your answers at <https://eksamen.dtu.dk/>. To hand in your answers, write them in the file `answers.txt` (this file is available from the same place you downloaded this file). When you are done, upload the `answers.txt` file (and nothing else). Double-check that you uploaded the correct version of the file from your computer.

Do not change the format of `answers.txt`

The file is automatically parsed after hand-in. Do not change the file format of `answers.txt` to any other format such as `rtf`, `docx`, or `pdf`. Do not change the file structure. Only edit the portions of the file indicated by question marks.

Attribute description		Abbrev.
x_1	Input voltage (Volt)	VIN
x_2	Input frequency (Hertz)	FIN
x_3	Amplification factor [0.10-2.30]	AMP
x_4	Power consumption (Watt)	PWR
x_5	Electromagnetic radiance (Watt per square meter)	EMR
y	Protection mode	PROT
y_r	Total distortion and noise	THDN

Table 1: The Electronic Device Quality dataset considered in this exam contains 900 measurements of an electronic device for amplifying an input signal. The classification task concerns the prediction of whether the device's protection mode is Activated ($y = 2$) or Deactivated ($y = 1$), while the regression task concerns the prediction of the total distortion and noise introduced by the amplification device, y_r .

	x_1	x_2	x_3	x_4	x_5
1 pct.	-1.19	-1.6	-1.76	-0.7	-1.21
5 pct.	-1.19	-1.36	-1.57	-0.7	-1.21
25 pct.	-1.11	-0.78	-0.84	-0.68	-0.87
50 pct.	-0.08	-0.13	0.03	-0.5	-0.23
75 pct.	1.13	0.65	0.89	0.31	0.7
95 pct.	1.21	1.77	1.48	2.43	1.93
99 pct.	1.21	2.59	1.48	3.21	2.25

Table 2: Percentiles for all attributes in the Electronic Device Quality dataset described in Table 1.

Question 1. We consider the standardized version of the Electronic Device Quality dataset described in Table 1, i.e., each attribute has a mean of zero and a standard deviation of one.

Figure 1 shows a histogram plot of two unidentified attributes A and B from the Electronic Device Quality dataset. In Table 2, specific percentiles of all the attributes are listed.

Which one of the following statements is true?

- A. Histogram A corresponds to attribute x_3 , and histogram B corresponds to attribute x_1 .
- B. Histogram A corresponds to attribute x_1 , and histogram B corresponds to attribute x_4 .
- C. Histogram A corresponds to attribute x_4 , and histogram B corresponds to attribute x_2 .
- D. Histogram A corresponds to attribute x_2 , and histogram B corresponds to attribute x_1 .
- E. Don't know.

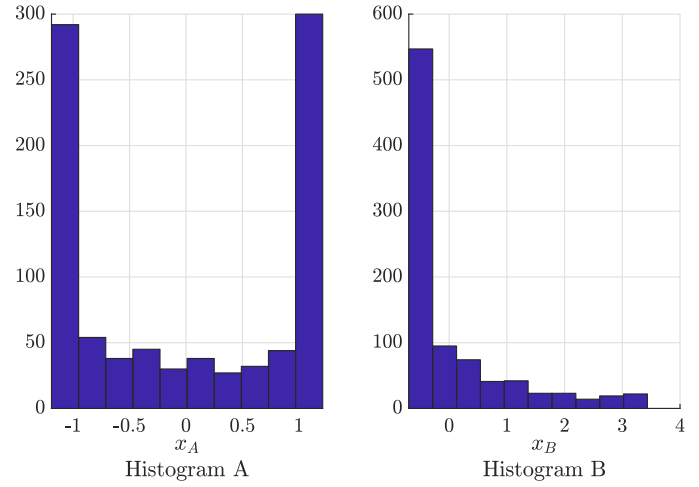


Figure 1: Histogram of two unidentified attributes (x_A and x_B) of the Electronic Device Quality dataset described in Table 1.

Question 2. Consider again the Electronic Device Quality dataset. The empirical covariance matrix of the first 4 attributes $x_1, \dots, x_4$ is given by:

$$\hat{\Sigma} = \begin{bmatrix} 1.0 & 0.036 & -0.077 & 0.643 \\ 0.036 & 1.0 & -0.03 & -0.029 \\ -0.077 & -0.03 & 1.0 & 0.459 \\ 0.643 & -0.029 & 0.459 & 1.0 \end{bmatrix}.$$

What is the empirical correlation of x_1 (VIN) and x_4 (PWR)?

- A. ≈ 0.643
- B. ≈ 0.8019
- C. ≈ -0.8019
- D. ≈ 0.4134
- E. Don't know.

Question 3. Consider a PCA analysis of the Electronic Device Quality dataset in the $N \times M$ matrix, $\mathbf{X}$, where $N = 900$ is the number of observations and $M = 5$ is the number of attributes, as usual. We standardize the $\mathbf{X}$ matrix column-wise by subtracting the mean and dividing by the standard deviation to obtain $\tilde{\mathbf{X}}$. With this matrix, we perform principal component analysis via singular value decomposition, obtaining $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \tilde{\mathbf{X}}$. The three matrices are found to be:

$$\mathbf{U} = \begin{bmatrix} 0.000 & -0.041 & -0.032 & -0.045 & -0.003 \\ -0.048 & 0.044 & -0.030 & -0.016 & 0.058 \\ 0.010 & -0.055 & -0.026 & -0.027 & -0.054 \\ -0.017 & -0.039 & 0.013 & -0.021 & -0.037 \\ -0.065 & 0.018 & -0.030 & 0.038 & -0.016 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1)$$

$$\mathbf{\Sigma} = \begin{bmatrix} 49.164 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 31.401 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 29.678 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 12.436 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 7.837 \end{bmatrix} \quad (2)$$

$$\mathbf{V} = \begin{bmatrix} -0.47 & 0.52 & -0.24 & 0.54 & 0.40 \\ 0.00 & 0.44 & 0.90 & -0.04 & 0.02 \\ -0.31 & -0.73 & 0.37 & 0.41 & 0.27 \\ -0.58 & -0.04 & -0.01 & -0.73 & 0.36 \\ -0.60 & 0.01 & 0.01 & 0.08 & -0.80 \end{bmatrix} \quad (3)$$

With the SVD result in hand, the aim is to obtain a principal component representation with the fewest possible components retaining at least 90 % of the total variance.

For the first observation in $\tilde{\mathbf{X}}$, $\tilde{\mathbf{x}}_1 = [1.21 \ 0.05 \ -1.43 \ -0.43 \ -0.59]^\top$, what would the coordinates, $\mathbf{b}_1$, be when $\tilde{\mathbf{x}}_1$ is projected onto the relevant subspace?

- A. $\mathbf{b}_1^\top = [0.021, -1.298, -0.953, -0.560]$
- B. Part of the first row in the matrix $\tilde{\mathbf{X}}\mathbf{V}$.
- C. $\mathbf{b}_1^\top = [0, -0.048, 0.010]$
- D. $\mathbf{b}_1^\top = [-0.466, -0.522, -0.243, -0.536]$
- E. Don't know.

Question 4. Consider again the PCA analysis for the Electronic Device Quality dataset, in particular the SVD decomposition of $\tilde{\mathbf{X}}$ in Equation (3). Which one of the following statements is true?

- A. An observation with a low value of x_1 (**VIN**), a low value of x_3 (**AMP**), a low value of x_4 (**PWR**), and a high value of x_5 (**EMR**) will typically have a positive value of the projection onto principal component number 5.
- B. An observation with a high value of x_1 (**VIN**), a high value of x_3 (**AMP**), a high value of x_4 (**PWR**), and a high value of x_5 (**EMR**) will typically have a positive value of the projection onto principal component number 1.
- C. An observation with a high value of x_1 (**VIN**), a high value of x_2 (**FIN**), and a low value of x_3 (**AMP**) will typically have a negative value of the projection onto principal component number 2.
- D. An observation with a low value of x_1 (**VIN**), a high value of x_2 (**FIN**), and a high value of x_3 (**AMP**) will typically have a positive value of the projection onto principal component number 3.
- E. Don't know.

Question 5. Consider the application of Kernel Density Estimation (KDE) to a synthetic 1D problem based on three observed data points $\mathbf{X} = \{-0.7, 1.1, 1.8\}$. Two possible kernel widths $\lambda = \{0.5, 1\}$ are considered with the final choice to be made based on leave-one-out cross-validation on $\mathbf{X}$.

Determine which one of the following test points is *most* likely to be a potential outlier using the KDE approach.

- A. $x^* = -0.9$
- B. $x^* = -0.3$
- C. $x^* = 0.6$
- D. $x^* = 1.2$
- E. Don't know.

	o_1	o_2	o_3	o_4	o_5	o_6	o_7	o_8	o_9	o_{10}
o_1	0.0	3.7	3.4	2.3	1.8	1.6	2.4	2.6	2.5	4.3
o_2	3.7	0.0	3.2	5.1	3.5	2.9	4.4	4.6	4.0	7.2
o_3	3.4	3.2	0.0	4.1	2.6	4.0	2.7	5.3	2.9	6.1
o_4	2.3	5.1	4.1	0.0	3.7	3.2	1.7	2.7	1.9	2.6
o_5	1.8	3.5	2.6	3.7	0.0	2.8	3.2	4.4	3.4	5.6
o_6	1.6	2.9	4.0	3.2	2.8	0.0	3.4	2.1	3.1	5.3
o_7	2.4	4.4	2.7	1.7	3.2	3.4	0.0	3.8	0.8	4.1
o_8	2.6	4.6	5.3	2.7	4.4	2.1	3.8	0.0	3.5	4.0
o_9	2.5	4.0	2.9	1.9	3.4	3.1	0.8	3.5	0.0	4.4
o_{10}	4.3	7.2	6.1	2.6	5.6	5.3	4.1	4.0	4.4	0.0

Table 3: The pairwise Euclidian distances, $d(o_i, o_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2 = \sqrt{\sum_{k=1}^M (x_{ik} - x_{jk})^2}$ between 10 observations from the Electronic Device Quality dataset (recall that $M = 5$). Each observation o_i corresponds to a row of the data matrix $\mathbf{X}$ of Table 1. The colors indicate classes such that the black observations $\{o_1, o_2, o_3, o_4, o_5, o_6, o_7\}$ belong to class C_1 (corresponding to Deactivated), and the red observations $\{o_8, o_9, o_{10}\}$ belong to class C_2 (corresponding to Activated). To avoid single features to dominate, the dataset was standardized by subtracting the mean and dividing by the standard deviation.

Question 6. To examine if observation o_1 may be an outlier, we will calculate the K -nearest neighborhood density using only the observations and distances in Table 3. For an observation o_i , recall the density is computed using the set of K nearest neighbors of observation o_i excluding the i 'th observation itself, $N_{\mathbf{X}_{\setminus i}}(o_i, K)$, and is denoted by $\text{density}_{\mathbf{X}_{\setminus i}}(o_i, K)$. What is the density for observation o_1 for $K = 3$ nearest neighbors?

- 0.625
- 0.462
- 1.139
- 0.526
- Don't know.

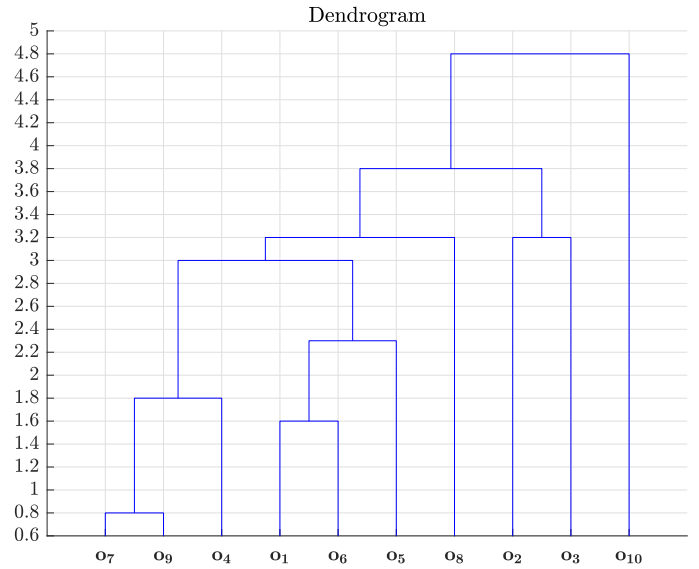


Figure 2: Dendrogram.

Question 7. Consider the observations and the pairwise distances in Table 3 along with the dendrogram in Figure 2.

Determine which linkage function was used to produce the dendrogram.

- Minimum linkage.
- Average linkage.
- Maximum linkage.
- The linkage function can not be determined from the available information.
- Don't know.

Question 8. Consider a Gaussian mixture model (GMM) with three components that define the density from which we draw random samples to create a dataset. We compute the principal components of this dataset. The corresponding directions are shown in Fig. 3. Which one of the following GMMs was used to generate the data?

A.

$$\begin{aligned}\mu_1 &= \begin{bmatrix} 4.9 \\ -2.6 \end{bmatrix} & \Sigma_1 &= \begin{bmatrix} 1.47 & 1.3 \\ 1.3 & 1.54 \end{bmatrix} & w_1 &= 0.34 \\ \mu_2 &= \begin{bmatrix} -0.04 \\ 2.6 \end{bmatrix} & \Sigma_2 &= \begin{bmatrix} 5.3 & -0.1 \\ -0.1 & 0.32 \end{bmatrix} & w_2 &= 0.12 \\ \mu_3 &= \begin{bmatrix} -1.17 \\ -1.13 \end{bmatrix} & \Sigma_3 &= \begin{bmatrix} 0.84 & 0.89 \\ 0.89 & 4.4 \end{bmatrix} & w_3 &= 0.54\end{aligned}$$

B.

$$\begin{aligned}\mu_1 &= \begin{bmatrix} 4.9 \\ -2.6 \end{bmatrix} & \Sigma_1 &= \begin{bmatrix} 0.84 & 0.89 \\ 0.89 & 4.4 \end{bmatrix} & w_1 &= 0.12 \\ \mu_2 &= \begin{bmatrix} -0.04 \\ 2.6 \end{bmatrix} & \Sigma_2 &= \begin{bmatrix} 1.47 & 1.3 \\ 1.3 & 1.54 \end{bmatrix} & w_2 &= 0.34 \\ \mu_3 &= \begin{bmatrix} -1.17 \\ -1.13 \end{bmatrix} & \Sigma_3 &= \begin{bmatrix} 5.3 & -0.1 \\ -0.1 & 0.32 \end{bmatrix} & w_3 &= 0.54\end{aligned}$$

C.

$$\begin{aligned}\mu_1 &= \begin{bmatrix} 4.9 \\ -2.6 \end{bmatrix} & \Sigma_1 &= \begin{bmatrix} 5.3 & -0.1 \\ -0.1 & 0.32 \end{bmatrix} & w_1 &= 0.34 \\ \mu_2 &= \begin{bmatrix} -0.04 \\ 2.6 \end{bmatrix} & \Sigma_2 &= \begin{bmatrix} 1.47 & 1.3 \\ 1.3 & 1.54 \end{bmatrix} & w_2 &= 0.12 \\ \mu_3 &= \begin{bmatrix} -1.17 \\ -1.13 \end{bmatrix} & \Sigma_3 &= \begin{bmatrix} 0.84 & 0.89 \\ 0.89 & 4.4 \end{bmatrix} & w_3 &= 0.54\end{aligned}$$

D.

$$\begin{aligned}\mu_1 &= \begin{bmatrix} 4.9 \\ -2.6 \end{bmatrix} & \Sigma_1 &= \begin{bmatrix} 5.3 & -0.1 \\ -0.1 & 0.32 \end{bmatrix} & w_1 &= 0.12 \\ \mu_2 &= \begin{bmatrix} -0.04 \\ 2.6 \end{bmatrix} & \Sigma_2 &= \begin{bmatrix} 1.47 & 1.3 \\ 1.3 & 1.54 \end{bmatrix} & w_2 &= 0.34 \\ \mu_3 &= \begin{bmatrix} -1.17 \\ -1.13 \end{bmatrix} & \Sigma_3 &= \begin{bmatrix} 0.84 & 0.89 \\ 0.89 & 4.4 \end{bmatrix} & w_3 &= 0.54\end{aligned}$$

E. Don't know.

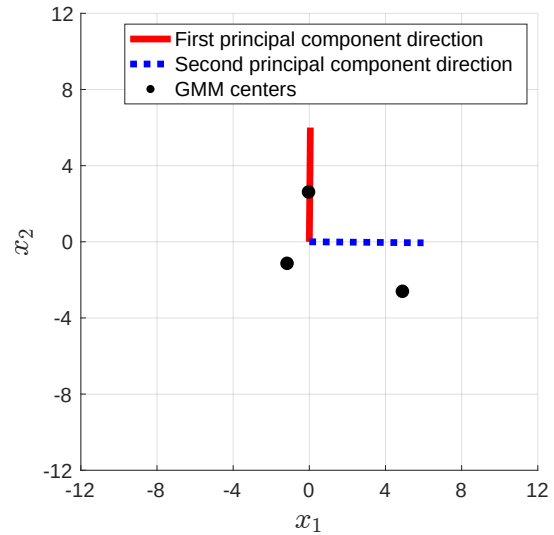


Figure 3: The two principal component directions illustrated along with the centers of the components in the Gaussian mixture model. The lines indicating the principal component directions have arbitrary lengths (for illustrative purposes).

Question 9. Consider again the Electronic Device Dataset. Suppose we wish to predict the class label y using a decision tree model. To improve performance, we apply AdaBoost (the specific variant described in Chapter 17 of the lecture notes). We will use the full dataset for training, and it will be denoted by $\mathcal{D}$. We run $T = 3$ boosting rounds, and suppose that the algorithm proceeds as follows:

- The first classifier, f_1 , has a weighted error, $\epsilon_{t=1}$, of 0.50 when evaluated on $\mathcal{D}$.
- The second classifier, f_2 , has a weighted error, $\epsilon_{t=2}$, of 0.75 when evaluated on $\mathcal{D}$.
- The third classifier, f_3 , has a weighted error, $\epsilon_{t=3}$, of 0.50 when evaluated on $\mathcal{D}$.

What is the *accuracy* of the combined classifier (majority voting classifier) produced by AdaBoost when evaluated on $\mathcal{D}$?

- A. ≈ 0.25
- B. ≈ 0.50
- C. ≈ 0.63
- D. ≈ 0.75
- E. Don't know.

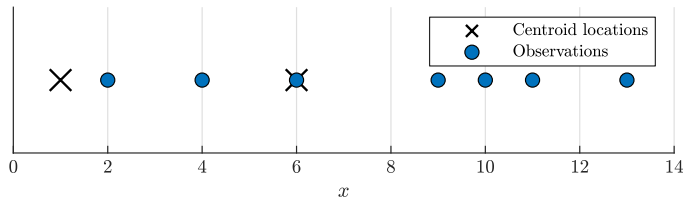


Figure 4: A small 1-dimensional dataset and initial values of centroids.

Question 10. Consider a small dataset comprised of $N = 7$ one-dimensional observations shown as the filled circles in Figure 4.

Suppose a k -means algorithm is applied to the dataset with $K = 2$ and using Euclidean distances. We will assume the location of the centroids are initialized to the values indicated by the crosses in Figure 4. After initialization, the k -means algorithm is evaluated for one step, comprised of assigning observations to centroids and updating the location of the centroids. After the first step, what will be the new location of the centroids?

- A. $\mu_1 = 3$, and $\mu_2 = \frac{49}{5}$.
- B. $\mu_1 = 4$, and $\mu_2 = \frac{43}{4}$.
- C. $\mu_1 = 2$, and $\mu_2 = \frac{53}{6}$.
- D. $\mu_1 = \frac{22}{3}$, and $\mu_2 = 11$.
- E. Don't know.

Question 11. Consider a regularized linear regression model trained using $\lambda = 2.23$ on the Electronic Device Quality dataset using the five attributes in addition to second-order polynomial terms as input, i.e.

$$\tilde{\mathbf{x}} = [1, x_1, x_2, \dots, x_5, x_1^2, x_2^2, \dots, x_5^2].$$

The model performs poorly on a held-out test set and has been confirmed to have very high bias and very low variance.

Which one of the following actions is most likely to improve the generalization performance on unseen

data?

- A. Increasing the amount of regularization applied to the model.
- B. Remove the second-order polynomial transformations from the input vector.
- C. Decrease the amount of regularization applied to the model.
- D. Train on the original training dataset with $\lambda = 2.23$ but only test on half the observations in the test set.
- E. Don't know.

Question 12. Suppose Hunt's algorithm is used to compute the purity gain for a decision tree (DT) classifier trained on the Electronic Device Quality dataset using a hold-out cross-validation procedure. The purity gain is defined as $\Delta = I(r) - \sum_{k=1}^K \frac{N(v_k)}{N(r)} I(v_k)$, where $I(z)$ is the impurity measure evaluated at the root, r , or a specific branch, v_k .

At the root, the training set contains 400 examples from class 1 and 100 from class 2. A particular split along x_4 results in two branches such that:

- Branch v_1 has 175 observations from class 1 and 75 observations from class 2.
- Branch v_2 has 225 observations from class 1 and 25 observations from class 2.

Determine which one of the following expressions correctly computes the purity gain for the provided split when using the *entropy* impurity measure.

- A. $\Delta = -\sum_{c=1}^2 p(c | r) \log_2 p(c | r) - \sum_{c=1}^2 p(c | v_1) \log_2 p(c | v_1)$
- B. $\Delta = \sum_{c=1}^2 (-p(c | r) \log_2 p(c | r) + \frac{1}{2} p(c | v_1) \log_2 p(c | v_1) + \frac{1}{2} p(c | v_2) \log_2 p(c | v_2))$
- C. $\Delta = -\sum_{c=1}^2 p(c | r) \log_2 p(c | r) + \sum_{k=1}^2 \sum_{c=1}^2 p(c | v_k) \log_2 p(c | v_k)$
- D. $\Delta = -\sum_{k=1}^2 \sum_{c=1}^2 p(c | v_k) \log_2 p(c | v_k)$
- E. Don't know.

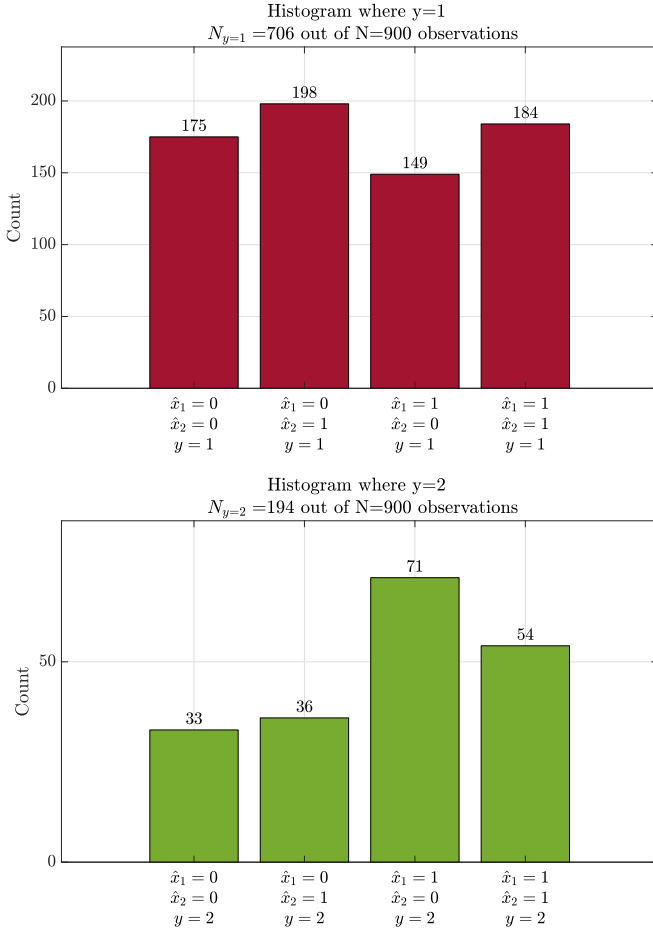


Figure 5: Histogram of the binarized version of the Electronic Device Quality dataset described in Table 1.

Question 13. Consider the Electronic Device Quality dataset from Table 1. The attributes have been binarized to produce new features denoted $\hat{x}_1, \hat{x}_2, \dots, \hat{x}_M$. A summary of the binarised dataset is given in Figure 5 for a subset of the (binarized) attributes and label, namely $\hat{x}_1, \hat{x}_2$ and y .

Determine the probability of $y = 1$ given an observation has $\hat{x}_1 = 0$ and $\hat{x}_2 = 1$ using Bayes theorem (without applying the Naïve Bayes assumption).

- A. 0.846
- B. 0.664
- C. 0.714
- D. 0.28
- E. Don't know.

Question 14. Consider clustering $N = 10$ observations using two clustering algorithms. The resulting cluster assignments is indicated by:

$$Q = [\begin{matrix} ? & ? & ? & 1 & 2 & 1 & 2 & 1 & 3 & 1 \end{matrix}]$$

$$Z = [\begin{matrix} ? & ? & ? & 2 & 3 & 3 & 4 & 2 & 1 & 1 \end{matrix}]$$

The elements in the vectors indicate the cluster index for a specific observation with “?” representing a missing/unknown cluster assignment.

We use the *Rand index* to compare the two results. It is known that the total number of times Q and Z agree that two observations are *not* in the same cluster is $D = 20$. Additionally, it is known that the total number of times Q and Z agree that two observations are in the same cluster is $S = 7$. Which one of the following options is the correct combination of missing assignments?

A.

$$Q = [\begin{matrix} 3 & 1 & 3 & 1 & 2 & 1 & 2 & 1 & 3 & 1 \end{matrix}]$$

$$Z = [\begin{matrix} 1 & 3 & 1 & 2 & 3 & 3 & 4 & 2 & 1 & 1 \end{matrix}]$$

B.

$$Q = [\begin{matrix} 1 & 1 & 2 & 1 & 2 & 1 & 2 & 1 & 3 & 1 \end{matrix}]$$

$$Z = [\begin{matrix} 4 & 1 & 3 & 2 & 3 & 3 & 4 & 2 & 1 & 1 \end{matrix}]$$

C.

$$Q = [\begin{matrix} 1 & 1 & 1 & 1 & 2 & 1 & 2 & 1 & 3 & 1 \end{matrix}]$$

$$Z = [\begin{matrix} 2 & 3 & 2 & 2 & 3 & 3 & 4 & 2 & 1 & 1 \end{matrix}]$$

D.

$$Q = [\begin{matrix} 1 & 2 & 2 & 1 & 2 & 1 & 2 & 1 & 3 & 1 \end{matrix}]$$

$$Z = [\begin{matrix} 1 & 3 & 4 & 2 & 3 & 3 & 4 & 2 & 1 & 1 \end{matrix}]$$

E. Don't know.

i	z_i^A	z_i^B	z_i^C
1	17	21	17
2	21	21	22
3	19	23	18
4	20	21	20
5	22	22	53

Table 4: Squared error from three regression models for $n = 5$ observations.

Question 15. Consider comparing three regression models (A, B, and C) using hold-out cross-validation and the setup presented in Section 11.3.6 of the lecture notes. The squared error on each of the five observations for all three models is shown in Table 4.

Determine which one of the following statements is correct.

- A. Model A's losses are always lower or equal to Model B's losses, thus a paired test on the difference in performance between the models is guaranteed to show that that Model A is better than Model B with a p -value < 0.05 .
- B. The confidence interval for the estimated difference in performance between Model A and Model C is $\approx [17.0, 53.0]$
- C. The losses of Model C seems far from being normally distributed due to the outlier ($z_5^C = 53$), so one should be careful with any application and interpretation of the usual paired test.
- D. When comparing Model A's and Model B's losses using a paired test, the degrees of freedom parameter ν in the test will be $\nu = (n - 1) \times n = 20$.
- E. Don't know.

i	1	2	3	4	5
$y_{r,i} - f(\hat{\mathbf{x}}_i, \hat{\mathbf{w}})$	0.00162	-0.00756	-0.0161	0.0248	-0.00274

Table 5: Regression result.

Question 16. Consider a regularized linear regression model on the (non-standardized) attribute, y_r , using three of the standardized attributes x_3, x_4, x_5 . We use a standard regression function defined as $f(\hat{\mathbf{x}}, \hat{\mathbf{w}}) = \hat{\mathbf{x}}^\top \hat{\mathbf{w}}$, where $\hat{\mathbf{w}}$ is the weight vector including the intercept term as the first element and $\hat{\mathbf{x}}^\top$ contains the standardized attributes augmented with a constant for the intercept term, i.e., $\hat{\mathbf{x}} = [1, \tilde{\mathbf{x}}]^\top$.

Consider an error function for the regularized linear regression problem defined as

$$E(\hat{\mathbf{w}}) = \frac{1}{N} \sum_{i=1}^N L(y_{r,i}, f(\hat{\mathbf{x}}_i, \hat{\mathbf{w}})) + \lambda \|\mathbf{w}\|_2^2,$$

where the function $L(y_{r,i}, \hat{y}_{r,i})$ is chosen to be the *squared* difference between the actual observation, $y_{r,i}$, and the prediction, $\hat{y}_{r,i}$. $\mathbf{w}$ is the weight vector *excluding* the intercept term.

The model parameters are estimated based on five observations with the relevant information shown in Table 5, resulting in $\hat{\mathbf{w}} = [0.0318, 0.04, 0.000197, w_5]^\top$ and $E(\hat{\mathbf{w}}) \approx 0.00025757$. $\lambda = 0.02$ was used in the estimation.

Determine which value of w_5 was estimated.

- A. $w_5 \approx -0.043$
- B. $w_5 \approx 0.018$
- C. $w_5 \approx 0.22$
- D. $w_5 \approx 0.60$
- E. Don't know.

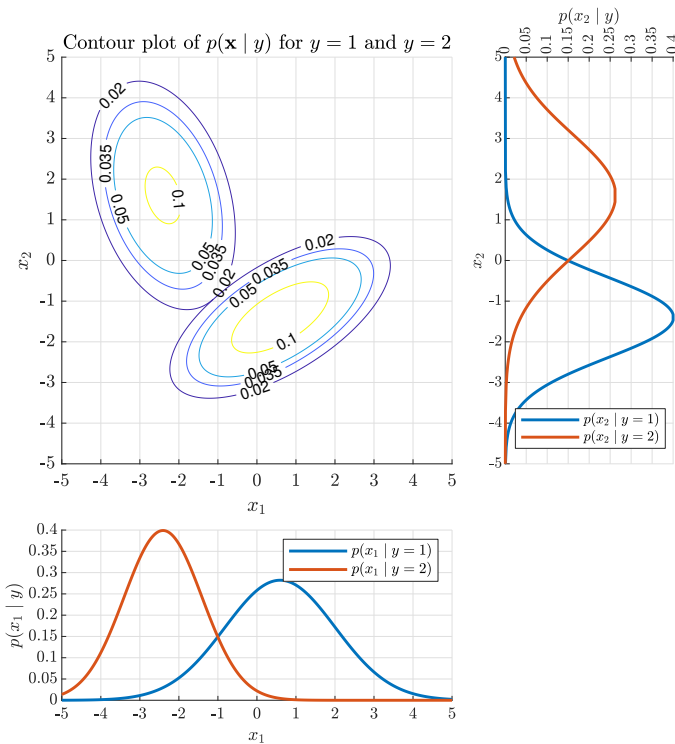


Figure 6: Conditional distributions related to a Naïve Bayes classifier.

Question 17. Consider the task of classifying the label y from two continuous attributes, namely x_1 and x_2 using Bayes rule.

We use a normal distribution to model the class conditional distribution of $\mathbf{x} = [x_1, x_2]^\top$, i.e. in general $p(\mathbf{x} | \mu_{y=k}, \Sigma_{y=k})$.

The specific distributions are illustrated in Figure 6. Additionally, it is noted that $p(y = 1) = 0.784$ and $p(y = 2) = 0.216$.

Determine the value of $p(y = 1 | \mathbf{x} = [-1, 0]^\top)$ using a Naïve Bayes classifier.

- A. ≈ 0.675
- B. ≈ 0.784
- C. ≈ 0.864
- D. ≈ 0.5
- E. Don't know.

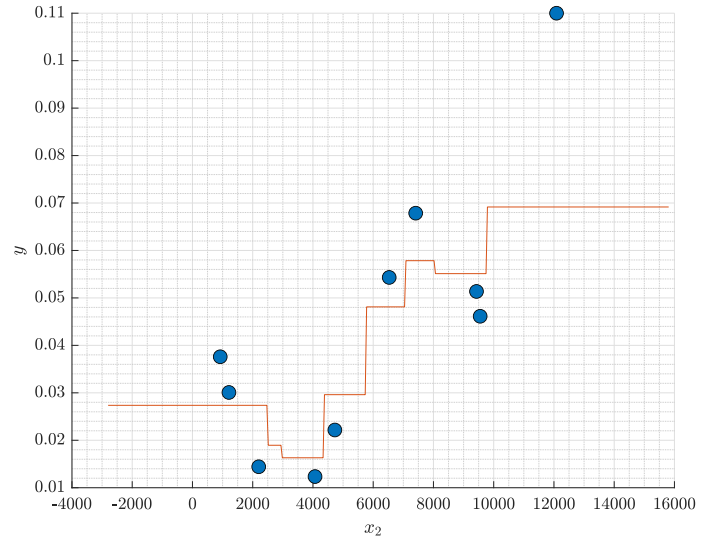


Figure 7: KNN regression model in which the red line is fitted to a small 1-dimensional dataset.

Question 18. Suppose a K -nearest neighbors regression model is fitted to a small 1-dimensional dataset with $N = 10$ observations. The predicted response is shown in Figure 7. How many neighbors (i.e. K) was used?

- A. $K = 5$
- B. $K = 2$
- C. $K = 3$
- D. $K = 4$
- E. Don't know.

Question 19. Consider again the Electronic Device Quality dataset in Table 1. We apply forward or backward selection to find an interpretable linear regression model which uses a subset of the $M = 5$ attributes to predict the y_r attribute.

Recall that backward and forward selection chooses models based on the test error as determined by cross-validation. In our case we use the hold-out method to generate a single test/training split.

Suppose backward selection ends up selecting the attributes x_1 and x_5 .

Determine which one of the following statements is correct.

- A. Forward selection is guaranteed to also select x_1 and x_5 .
- B. Backward selection will always result in fewer attributes being selected compared to forward selection.
- C. Backward selection requires *testing* 13 models if the procedure selects x_1 and x_5 .
- D. Forward selection requires *training* 13 models if the procedure selects two attributes.
- E. Don't know

Question 20. Suppose we fit three different logistic regression models (A, B and C) on the Electronic Device Quality dataset.

Model A uses x_1 .

Model B uses x_1 and x_2 .

Model C uses all five attributes $x_1, \dots, x_5$.

We standardize the input attributes and add an intercept column with ones in front of the attribute columns.

The estimated parameters/weights are:

$$\mathbf{w}^A = [-1.332, 0.378]^\top$$

$$\mathbf{w}^B = [-1.346, 0.387, -0.222]^\top$$

$$\mathbf{w}^C = [-15.913, 9.028, -0.273, 18.852, 2.133, -5.583]^\top$$

where the first element in each vector corresponds to the intercept/bias term.

We consider the (already standardized) observation $\mathbf{x}_3 = [1.209, -0.303, -1.070, -0.459, 0.013]^\top$ and the prediction $\hat{y}_i = \sigma(\mathbf{x}_i^\top \mathbf{w})$ where $\sigma(z)$ is the logistic sigmoid function.

Determine which one of the following statements is correct.

- A. Model A's prediction is $\hat{y}_3^A \approx -0.87$
- B. Model A's prediction is $\hat{y}_3^A \approx 0.15$
- C. Model B's prediction is $\hat{y}_3^B \approx 0.31$
- D. It is not possible to make a prediction using $\mathbf{w}^C$ as the dimension of $\mathbf{w}^C$ does not match the input.
- E. Don't know.

DT		Predicted	
		Activated	Deactivated
Actual	Activated	373	110
	Deactivated	82	335

ANN		Predicted	
		Activated	Deactivated
Actual	Activated	355	128
	Deactivated	64	353

Table 6: Confusion matrices.

Question 21. Table 6 shows the confusion matrices related to a neural network (ANN) classifier and a decision tree (DT) classifier trained on a particular instance of the Electronic Device Quality dataset to predict y (PROT). The dataset has been resampled to almost balance the number of observations from the two classes, but the number of observations is still $N = 900$.

We define $y = 1$ to mean a *negative* outcome (i.e., PROT is Deactivated), and $y = 2$ to be a *positive* outcome (i.e., PROT is Activated).

On the particular instance of the dataset, the goal is to have low error rate and simultaneously avoid misclassifying a device as having its protection mode Deactivated when it is actually Activated.

Determine which one of the following statements is true.

- A. It is not possible to determine which classification model is better based on the provided information.
- B. The DT is preferred over the ANN because the DT's precision is higher than the ANN's precision.
- C. The DT is preferred over the ANN because the DT's false positive rate is lower than the ANN's false positive rate.
- D. The DT is preferred over the ANN because the DT's recall is higher than the ANN's recall.
- E. Don't know.

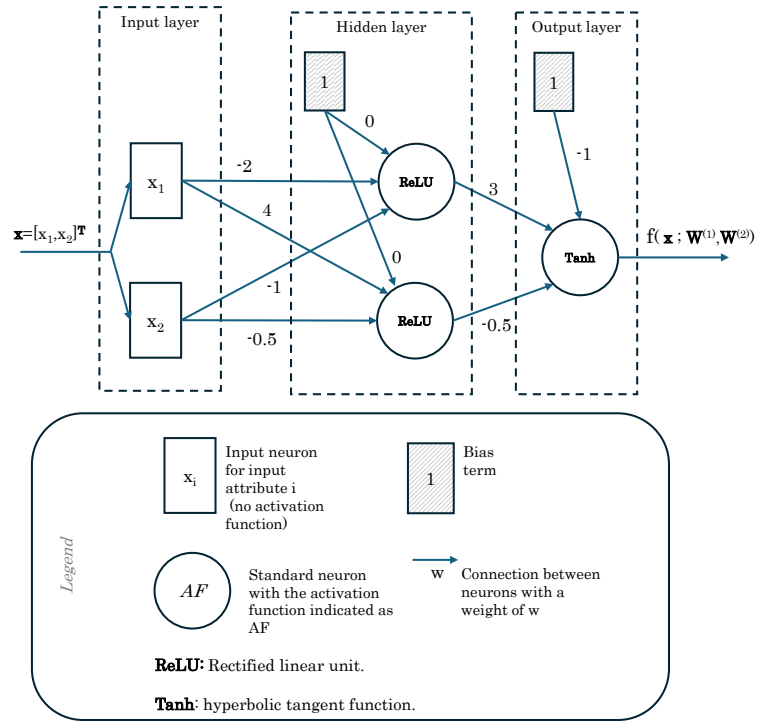


Figure 8: ANN schematic.

Question 22. Consider a two layer neural network $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ for regression with two hidden units (i.e. $\mathbf{z}^{(1)} \in \mathbb{R}^2$) and of the form

$$\mathbf{z}^{(1)} = h^{(1)}((\mathbf{W}^{(1)})^\top \tilde{\mathbf{x}}),$$

$$f(\mathbf{x}; \mathbf{W}^{(1)}, \mathbf{W}^{(2)}) = h^{(2)}((\mathbf{W}^{(2)})^\top \tilde{\mathbf{z}}^{(1)}),$$

with a schematic provided in Figure 8.

Given an input, $\mathbf{x} = [x_1 \ x_2]^\top = [0 \ -1]^\top$ determine the corresponding output.

- A. ≈ -1.00
- B. ≈ -0.85
- C. ≈ 0.94
- D. ≈ 1.75
- E. Don't know.

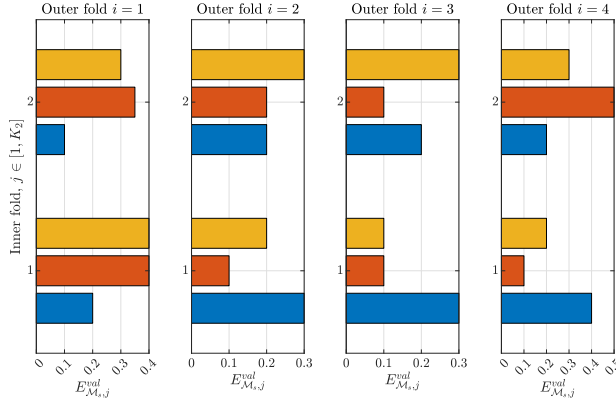
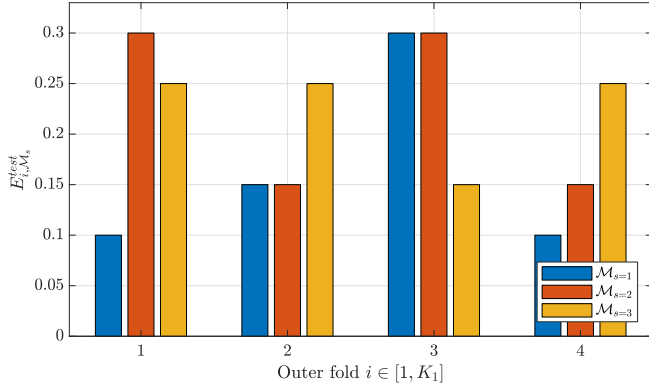


Figure 9: Result of running two-layer cross-validation.

Question 23. Consider evaluating three regression models ($\mathcal{M}_{s=1}$, $\mathcal{M}_{s=2}$ and $\mathcal{M}_{s=3}$), based on a dataset with 1200 observations. We use two-layer cross-validation (c.f. Algorithm 6 in the lecture notes) with $K_1 = 4$ outer folds and $K_2 = 2$ inner folds. The error is computed using the mean squared error.

The result of the procedure is shown in Figure 9. $E_{i, \mathcal{M}_s}^{test}$ denotes the performance of the models with complexities $s = \{1, 2, 3\}$ on the test set in the outer fold. Hence, we do not just evaluate the optimal model, $\mathcal{M}_{s^*}$, chosen based on the result of the inner fold, but compute the error for all three models on the test set in each of the outer folds.

Determine the value of the estimated generalization error $\hat{E}^{gen}$.

- A. $\hat{E}^{gen} \approx 0.2$
- B. $\hat{E}^{gen} \approx 0.3$
- C. $\hat{E}^{gen} \approx 0.1$
- D. $\hat{E}^{gen} \approx 0.25$
- E. Don't know.

	f_1	f_2	f_3	f_4	f_5
o_1	1	1	0	1	1
o_2	0	1	0	1	0
o_3	0	0	0	1	0
o_4	1	0	1	1	1
o_5	1	0	0	1	0
o_6	1	1	0	1	1
o_7	0	0	1	1	1
o_8	1	1	1	1	1
o_9	0	0	1	1	1
o_{10}	1	0	1	1	1

Table 7: Binarized version of the Electronic Device Quality dataset. Each of the features f_i are obtained by taking a feature x_i and letting $f_i = 1$ correspond to a value x_i greater than the median (otherwise $f_i = 0$). The colors indicate classes such that the black observations $\{o_1, o_2, o_3, o_4, o_5, o_6, o_7\}$ belong to class C_1 (corresponding to Deactivated), and the red observations $\{o_8, o_9, o_{10}\}$ belong to class C_2 (corresponding to Activated).

Question 24. We consider the binary matrix from Table 7 as a market basket problem consisting of $N = 10$ transactions $o_1, \dots, o_{10}$ and $M = 5$ items $f_1, \dots, f_5$. What is the *confidence* of the rule $\{f_2, f_3, f_4\} \rightarrow \{f_1, f_5\}$?

- A. The confidence is $\frac{3}{10}$
- B. The confidence is 1
- C. The confidence is $\frac{1}{5}$
- D. The confidence is $\frac{1}{10}$
- E. Don't know.

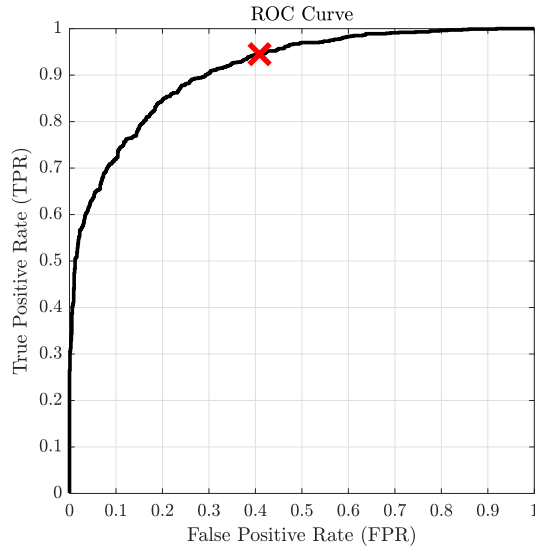


Figure 10: ROC.

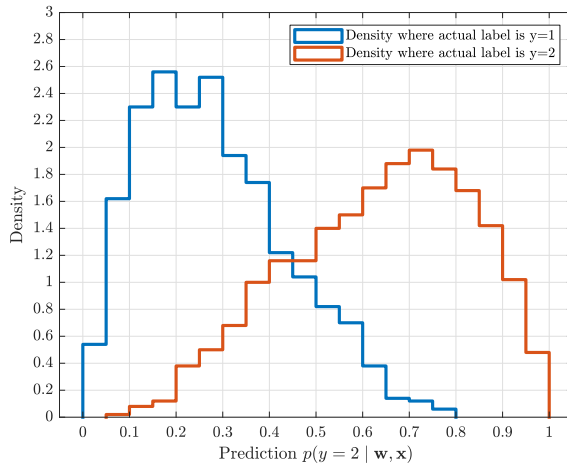


Figure 11: Density of predicted probabilities.

Question 25. Consider the ROC curve shown in Figure 10 originating from evaluating a logistic regression model.

Based on the density of the predictions shown in Figure 11, determine which threshold corresponds to the (FPR, TPR) -point indicated by the red cross in Figure 10.

- A. ≈ 0.01
- B. ≈ 0.1
- C. ≈ 0.3
- D. ≈ 0.7
- E. Don't know.

Question 26. Consider two classifiers evaluated on the Electronic Device Quality dataset with $(N = 900)$ observations:

Model A: A logistic regression model with the complexity controlled by the regularization parameter λ .

Model B: A neural network with the complexity controlled by the number of hidden layers.

The models were compared using on two-layer cross-validation (Algorithm 6 in the lecture notes) with $K_1 = 10$ outer folds and $K_2 = 3$ inner folds, and a correct/ideal application of McNemar's test.

McNemar's test resulted in an estimated difference in accuracy of $\hat{\theta} = -0.097$ along with an undisclosed confidence interval and p -value.

McNemar's test relies on a 2×2 matched-pair matrix, $\begin{bmatrix} n_{11} & n_{12} \\ n_{21} & n_{22} \end{bmatrix}$, where

- n_{11} : Number of times both classifiers are correct
- n_{12} : Number of times A is correct and B is wrong
- n_{21} : Number of times A is wrong and B is correct
- n_{22} : Number of times both classifiers are wrong

Determine which one of the following matrices was used in the estimation.

- A. $\begin{bmatrix} 707 & 13 \\ 100 & 80 \end{bmatrix}$
- B. $\begin{bmatrix} 71 & 1 \\ 10 & 8 \end{bmatrix}$
- C. $\begin{bmatrix} 236 & 4 \\ 33 & 27 \end{bmatrix}$
- D. $\begin{bmatrix} 77 & 104 \\ 17 & 702 \end{bmatrix}$

E. Don't know.

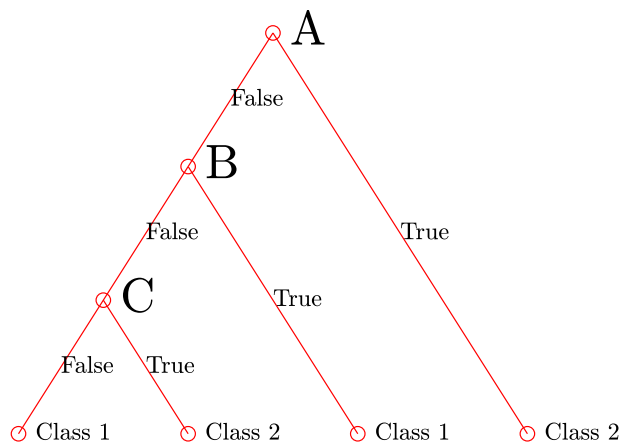


Figure 12: Example classification tree.

Question 27. We consider an artificial dataset of $N = 4000$ observations. The dataset is classified according to a decision tree of the form shown in Figure 12 resulting in a partition into classes indicated by the colors/markers in Figure 13. What is the correct rule assignment to the nodes in the decision tree?

A. $A: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 6 \\ 6 \end{bmatrix} \right\|_2 < 3, B: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\|_1 < 2,$
 $C: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\|_1 < 2$

B. $A: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\|_1 < 2, B: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 6 \\ 6 \end{bmatrix} \right\|_2 < 3,$
 $C: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\|_1 < 2$

C. $A: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 6 \\ 6 \end{bmatrix} \right\|_2 < 3, B: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\|_1 < 2,$
 $C: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\|_1 < 2$

D. $A: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\|_1 < 2, B: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\|_1 < 2,$
 $C: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 6 \\ 6 \end{bmatrix} \right\|_2 < 3$

E. Don't know.

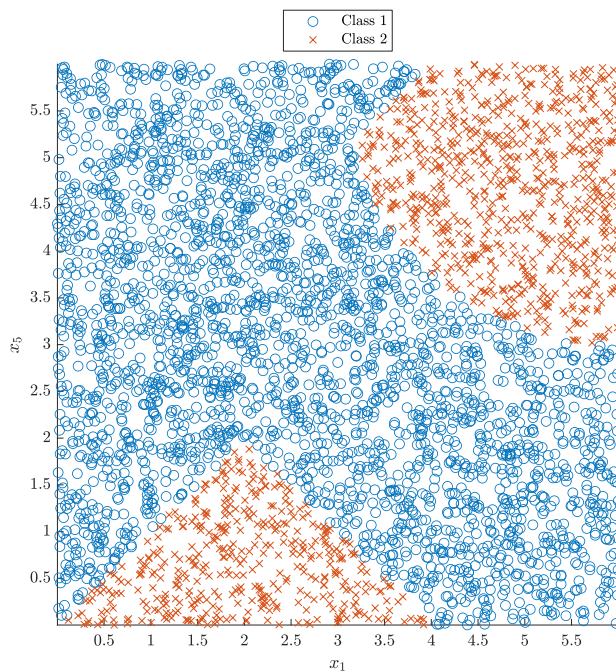


Figure 13: Classification boundary.